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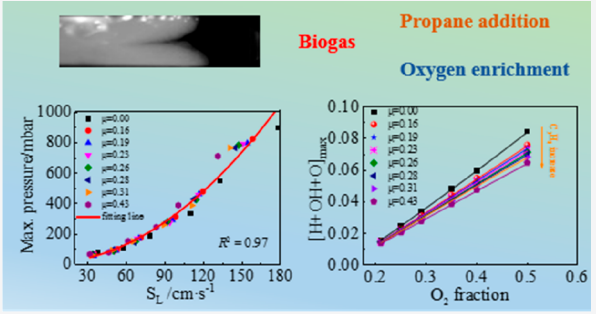
Effect of Propane Addition and Oxygen Enrichment on the Flame Characteristics of Biogas

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ABSTRACT: Mixing biogas with other fuels or increasing the oxygen fraction in the oxidizer can improve the combustion performance of biogas. In this study, we investigated the effects of propane addition and oxygen enrichment on the flame characteristics of biogas with a constant heating value in a half-open duct. The evolution of the flame structure revealed that the tulip structure took on two types of patterns at the oxygen fraction θ = 0.21. However, there was no salient tulip flame formation when θ > 0.21. Oxygen enrichment significantly increased the flame propagation velocity (V) and explosion pressure (P). Propane addition significantly reduced V_{max} at θ = 0.25 and 0.29. The maximum rate of pressure rise $[(dP/dt)_{max}]$ exhibited a nonlinear dependence on the maximum flame velocity (V_{max}). A similar trend was observed for the dependence of $(dP/dt)_{max}$ on the laminar burning velocity (LBV). Therefore, the LBV is an important parameter to predict V_{max} and $(dP/dt)_{max}$ values of mixtures. Furthermore, by using the PREMIX code and the UC San Diego mechanism, we simulated laminar flame burning properties to reveal the macroscopic flame propagation characteristics. The rate of propane consumption was higher than the rate of methane consumption. Additionally, oxygen enrichment played an important role in the entire reaction process, leading to an increase in the molar concentrations of microscopic free radicals and a promotion of the rates of production (ROP) of free radicals. In contrast, the addition of propane decreased the molar concentrations of microscopic free radicals and inhibited the ROP of free radicals.



1. INTRODUCTION

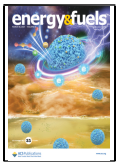
Currently, more than 85% of the world's energy demands is met by fossil fuels.[1] The emission of CO_2 and NO_x is increasing continuously, which has led to a significant greenhouse effect and air pollution.[2] Growing concerns regarding the amount of these gases emitted into the environment and predictions of the exhaustion of petroleum-derived fuels have motivated the development of clean energy.[3]

Biogas is produced through bacterial conversion of organic matter under anaerobic conditions and is a mixture of CH_4 and CO_2, with small amount of H_2S, N_2, CO, and O_2.[3,4] Compared with traditional fossil fuels, biogas has the advantages of abundant availability and lower pollution.[3] It is also very appropriate for power generation, heating systems, and other applications.[4] Interest in the use of biogas as an increasingly renewable and sustainable energy source has been promoted recently.[5] Dupont and Accorsi[6] investigated biogas explosion characteristics at various temperatures (30−70 °C) and at atmospheric pressure. Anggono et al.[7] determined the flammability limits of biogas−air mixtures using spherically expanding flames propagating. Ratna Kishore et al.[8] studied the adiabatic burning velocities for natural-gas-like mixtures over a range of equivalence ratios at atmospheric pressure. Hinton and Stone[9] measured the laminar burning velocity (LBV) of biogas over a wide range of temperatures and pressures. Zeng et al.[10] investigated the main factors influencing the combustion stability and LBV of biogas in laminar burning inside a constant-volume combustion chamber using a schlieren system.

However, biogas has an extremely low energy density, low propagation velocity, and narrower flammability ranges based on large amounts of inert gas (i.e., CO_2).[11] These characteristics have hindered the popularization of biogas. On the basis of these shortcomings, two methods have been developed to improve the combustion performance of biogas: mixing biogas with other fuels with superior characteristics or increasing the oxygen fraction in the oxidizer.[12] To evaluate both methods, this study raises the possibility of investigating the effect of propane addition and oxygen enrichment on the explosion characteristics of biogas.

A well-established alternative to gasoline is liquefied petroleum gas, where propane is a major component.[13−15]

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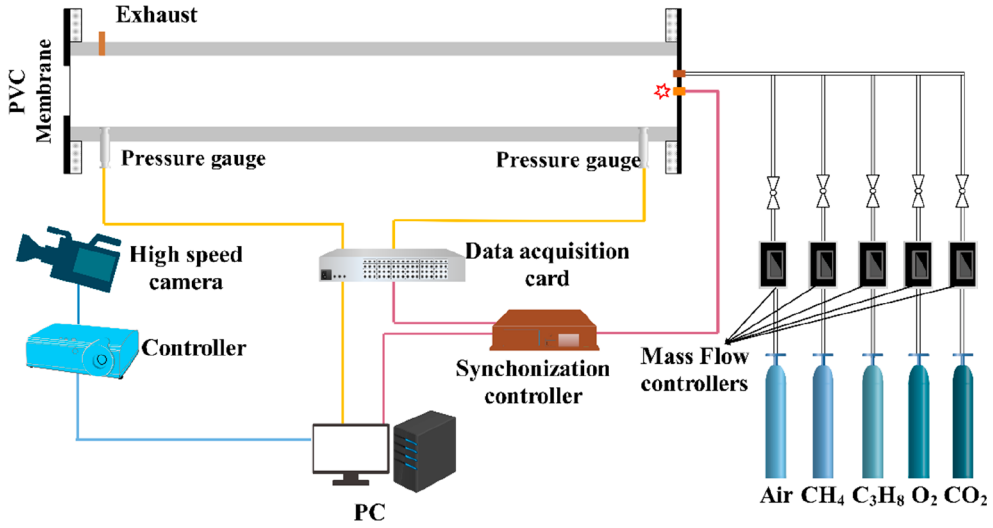


Figure 1. Schematic diagram of the experimental system.

The lower heating value of propane is 83.470 MJ/m^3, which is 2.33 times that of methane (35.823 MJ/m^3).[16] Propane has been mixed with biogas to improve its combustion properties in previous studies.[17−19] Law[20] and Kwon demonstrated that propane addition reduces the knocking occurrence of an internal combustion engine. Schuh et al.[21] demonstrated that propane addition reduces the ignition delay time of the methane−propane blends. Gómez Montoya et al.[22] pointed out that biogas mixtures with propane yield increased output power and generation efficiency for spark ignition biogas engines. Lee et al.[23] determined the burning velocities of biogas and biogas/propane mixtures. Luo et al.[24] measured the upper and lower flammability limits and explosive risk values for different mixtures of CH_4 and C_3H_8. Petersen et al.[18] investigated the ignition and oxidation kinetics of CH_4−C_3H_8 mixtures at gas turbine pressures by shock tube experiments and chemical kinetic modeling. Cardona and Amell[12] determined that biogas/propane/hydrogen mixtures have higher a LBV than pure biogas at 0.828 atm and 298 K while varying the air/fuel ratio.

Oxygen-rich combustion technology[25,26] can improve the combustion efficiency of biogas[12,22,27] and reduce CO_2 and NO_x emissions.[1,28,29] The concept of mixing these fuels is not new. However, compared to the study of a single fuel such as biogas and propane, there are relatively few published studies reporting the influence of propane addition and oxygen enrichment on the explosion characteristics of biogas. The increasing applications of biogas will introduce additional safety issues.[25,30,31] In addition, the typical phenomenon of combustion-induced rapid phase transition (CRPT) of a closed-vessel explosion can induce anomalous impulses in the pressure history in oxygen-enriched air.[25,31] Benedetto et al.[32,33] indicated that the CRPT can be mitigated by the presence of CO_2.

In real workplaces, biogas can be leaked into a confined space of large length-to-diameter ratios, such as tunnels, passageways, pipelines, silos, and mines, forming an explosion of any concentration of fuel−air mixtures within the explosion limits.[4,30] There is little knowledge about the flame propagation and pressure dynamics of biogas explosions in confined spaces that have a similar geometry of large length-to-width ratios. Furthermore, compared with freely propagating flames, the premixed flame will behave in a much more complicated manner when the flame propagates in a duct of large length-to-width ratio. In a sufficiently long, smooth duct, the laminar flame will undergo a series of structural changes. The most interesting and attractive flame shape is a tulip flame.[34−36] Furthermore, with propane addition and oxygen enrichment of biogas, complex fuel compositions can lead to unpredictable combustion performance.

Overall, this work aimed to study the effects of propane addition and oxygen enrichment on the flame characteristics of biogas experimentally in a half-open rectangular duct over a wide range of mixture compositions while the heating value of the blended fuel remained constant. This study provides experimental data and theoretical guidance for the combustion of biogas containing C_3H_8 and O_2 in a half-open duct. It is important to determine the flame acceleration mechanisms behind the deflagration-to-detonation transition, which are relevant for safety in transport pipelines, silos, and mines.

2. EXPERIMRNTAL SETUP

A schematic diagram of the experimental system is shown in Figure 1, similar to those of the previous studies.[37,38] All experiments were conducted in a transparent plexiglass explosion duct with a cross-sectional area of 100 × 100 mm^2, length of 1000 mm, and volume of 10 L, which can withstand the pressure of 2 MPa. The left end of the duct was sealed with a PVC film as a relief vent, and the opposite end was sealed with rubber pads and TP304 steel plates. A pair of spark electrodes was centrally flush with the steel plate at the right end of the duct, and the ignition voltage was 6 V. The transient flame development process was recorded by a high-speed camera (Dantec Corp.). Two pressure transducers (Shanghai MingKong Sensor Technology Ltd.) were employed to collect the dynamic pressure details in the duct. One pressure transducer installed at the bottom sidewall was 10 cm away from the open end, and another was installed on the steel plate at the right end of the duct. The acquisition frequency of the data acquisition card (model USB-1608Plus) was 15 kHz. In addition, a synchronization controller was used to ensure synchronous operation of the pressure transducers, the high-speed camera, and the ignitor, which facilitates coupling the analysis of flame development and pressure history.

The mixture consisted of CH_4, C_3H_8, O_2, CO_2, and dry air. The purity of CH_4, C_3H_8, O_2, and CO_2 were 99.995%, 99.99%, 99.995%, and 99.99%, respectively. The air is compressed air, and the ratio of the nitrogen to oxygen volume is 79:21. All the gases except air were produced by Jining Xieli Special Gas Co., Ltd. Five mass flow

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controllers (Alicat Corp.) were used to regulate the mass flow rates of individual gases at atmospheric pressure. The mass flow controllers have an accuracy of ±0.4% of the full scale. In order to minimize the uncertainty, all of the mass flow controllers were used above 10% and below 90% of their respective ranges. The various gases were premixed in a sufficient length of rubber tube. Then, the aeration process sustained 5 min through an inlet installed on the right steel plate. The mixture was discharged through an exhaust on the top wall of the duct close to the open end. Ten minutes elapsed before the ignition to ensure the homogeneity and stillness of the mixtures.

Initial experimental conditions are shown in Table 1. The oxygen fraction (θ) in the oxidizer is defined as $O_2/(O_2 + N_2)$. The propane fraction (μ) in the biogas/propane fuel and the methane fraction (γ) in the biogas are defined by eqs 1 and 2

Table 1. Initial Experimental Conditions

item	values
vessel shape	rectangular, 0.1 × 0.1 × 1.0 (m^3)
mixture	$CH_4/C_3H_8/O_2/CO_2/N_2$
initial pressure	1.0 atm
initial temperature	308.15 ± 2 K
equivalence ratio	1.0
oxygen fraction in oxidizer (θ)	0.21, 0.25, 0.29, 0.35, 0.40, and 0.50
propane fraction in fuel (μ)	0.00, 0.16, 0.19, 0.23, 0.26, 0.28, 0.31, and 0.43

$$\mu = \frac{V_{C_3H_8}}{V_{C_3H_8} + V_{biogas}} = \frac{V_{C_3H_8}}{V_{C_3H_8} + V_{CH_4} + V_{CO_2}} \quad (1)$$

$$\gamma = \frac{V_{CH_4}}{V_{CH_4} + V_{CO_2}} \quad (2)$$

where $V_{C_3H_8}$, V_{CH_4}, and V_{CO_2} were the volumes of C_3H_8, CH_4, and CO_2, respectively.

To keep the heating value of the biogas/propane fuel constant, μ and γ satisfied

$$(1 - \mu)\gamma \times 35.823 + \mu \times 83.47 = 35.823\ MJ/m^3 \quad (3)$$

To ensure the accuracy and repeatability, each composition was tested at least three times. The relative standard error (averaged in all cases) in experimental P_{max} was 2.4%.

3. RESULTS AND DISCUSSSION

3.1. Flame Structure Characteristics. Figure 2 presents the evolution of the flame structure over time for each combination of oxygen fraction θ and propane fraction μ. When the oxygen fractions θ are 0.35, 0.40, and 0.50, the flame structural evolution is similar to that for θ = 0.29. Therefore, they are not represented in Figure 2.

As shown in Figure 2a, when the oxygen fraction is θ = 0.21, a typical tulip flame appears; i.e., the flame convex toward the unburned gas changes to one convex toward the burned gas.[35,39] In the case of μ = 0.43 in Figure 2a, a salient tulip flame can be observed. The flame gradually evolves from a hemisphere (t = 20 ms) to a finger shape (t = 40 ms), a flat shape (t = 75 ms), and a tulip flame (t = 95 ms). As θ increases to 0.25, the reactivity of the mixture is enhanced. The tulip flame became distorted and the flame surface became progressively wrinkled. As θ increases from 0.25 to 0.29, only hemispherical and finger flames form, the flame front becomes more slender, and the luminance of the flame increases. The flames quickly pass out of the duct without forming a tulip flame, as shown in Figure 2c.

Zheng et al.[38] found that depending on oxygen fraction and equivalence ratio, the tulip structure of $CH_4/O_2/CO_2$ mixtures takes on three types of patterns. Alabaş et al.[40] indicated that compared to the combustion of pure methane with air, increasing the oxygen enrichment to 24% increases the combustion stability, but the stability decreases again at 26% oxygen enrichment under acoustic enforcement. What is more, oxygen enrichment has a significant influence on flame structure evolution in Figure 2. In addition, for θ = 0.21, when increasing the propane fraction μ from 0.00 to 0.43, there are different evolutionary processes for the tulip flame. Specifically, the evolution times of the different flame stages are extended. However, for θ > 0.21, there are no obvious changes in the flame structure as μ increases.

For θ = 0.21, the tulip structure takes on two types of patterns with varying propane fraction. The first type is presented in Figure 3a, where one can see a classic tulip flame at 76 ms. Subsequently, the two tongues of the flame front continue to convex toward the unburned gas. It is obvious that the upper tongue travels faster than the lower tongue (see the flame images at 88, 89, and 92 ms) in Figure 3a. The position of the cusp that sags into the burned gas remains stationary relative to the closed end of the duct, as indicated by the red line in Figure 3a. Therefore, the more slender upper tongue protrudes forward as if the flame is floating above. This asymmetric flame propagation pattern could be attributed to buoyancy.[2,9,10] Hinton and Stone[9] demonstrated that the buoyancy has a significant impact on the flame propagation when the LBV is less than 15 cm/s. As shown in Figure 4, all of the LBVs were greater than 30 cm/s in this study. Therefore, buoyancy is not the origin of the asymmetric flame tongues. In this set of experiment, the PVC film first ruptured in the upper section, resulting in a stronger gas flow in the upper section than in the lower section, which has a greater impact on the upper tongue. The second type of tulip flame formation is presented for μ = 0.43 in Figure 3b. The flame structure develops into a classic tulip at 97 ms, and then the upper and lower tongues of the flame front elongate toward the unburned gas in a symmetric manner. Compared to the cusp position at θ = 0.21 and μ = 0.00, the cusp position at θ = 0.21 and μ = 0.19 moves forward (see the flame images at t = 97, 110, and 117 ms that are marked by red lines). Additionally, the upper and lower tongues become thicker, move toward the duct axis, and then gradually merge into each other.

The flat flame is the signature of the formation of the tulip flame (the transition of the flame curvature from positive to negative signifies the occurrence of the tulip flame). As shown in Figure 2, the flat flame can only be observed at θ = 0.21 and 0.25. As the propane fraction μ increases, the formation distance of the flat flame gradually decreases (the ignition end is the zero point on the coordinate axis), as shown in Figure 5. As θ increases from 0.21 to 0.25, the formation distance of the flat flame clearly increases. Additionally, there is a consistent increase in the formation time of a flat flame when μ increases (from 61 to 75 ms for θ = 0.21 and from 31 to 46 ms for θ = 0.25). Di Benedetto et al.[12,45,46] demonstrated that oxygen enrichment increases the reactivity of a mixture, thereby increasing the LBV. The LBV increases with increasing θ, but it decreases with an increase in μ, as shown in Figure 4. For a fixed duct length, higher reactivity results in a shorter flame residence time and a longer flat flame formation distance. Thereafter, as the fuel reactivity continues to increase, the

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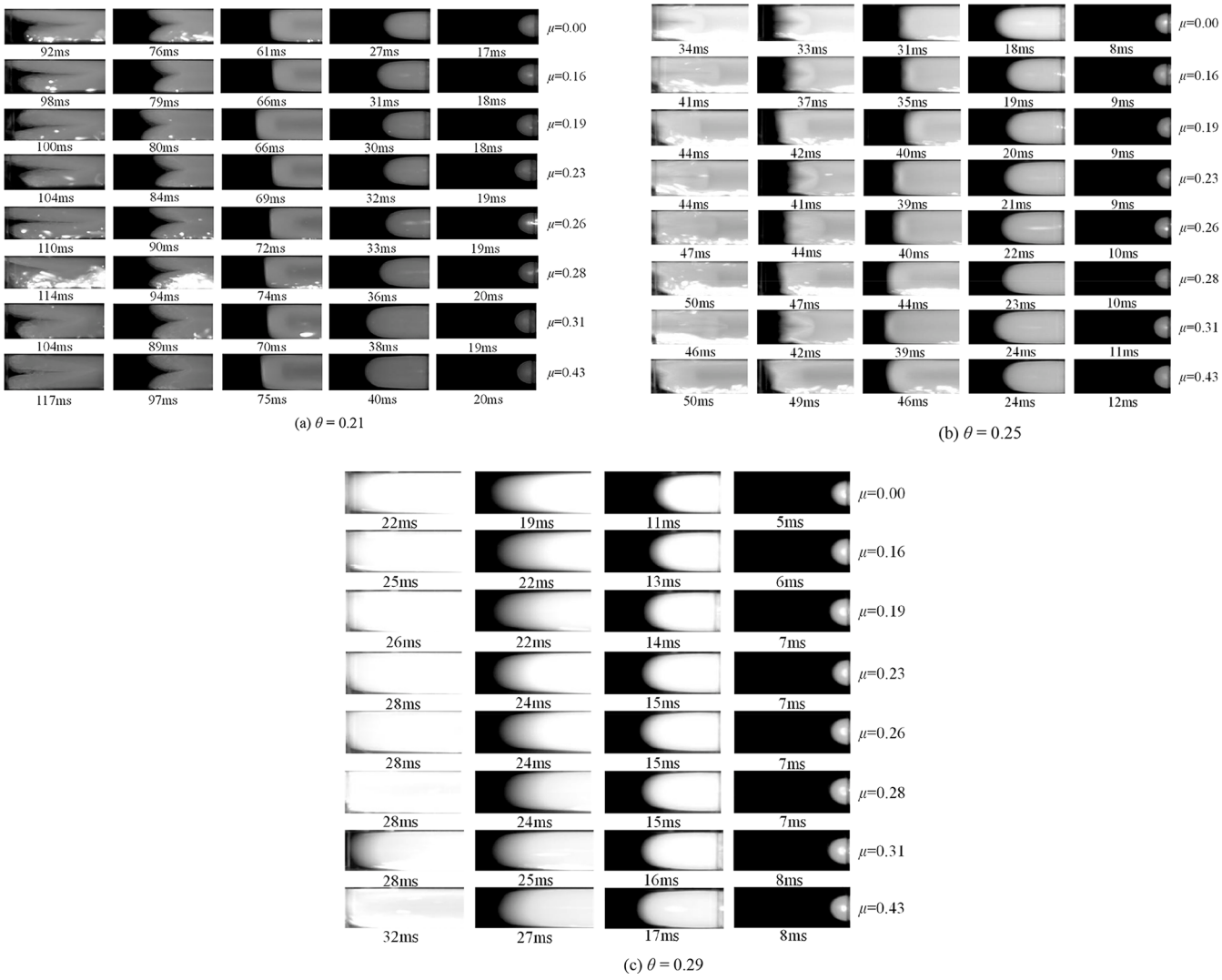


Figure 2. Flame propagation processes with different propane and oxygen fractions: (a) $\theta = 0.21$, (b) $\theta = 0.25$, (c) $\theta = 0.29$. Flame propagates from right to left.

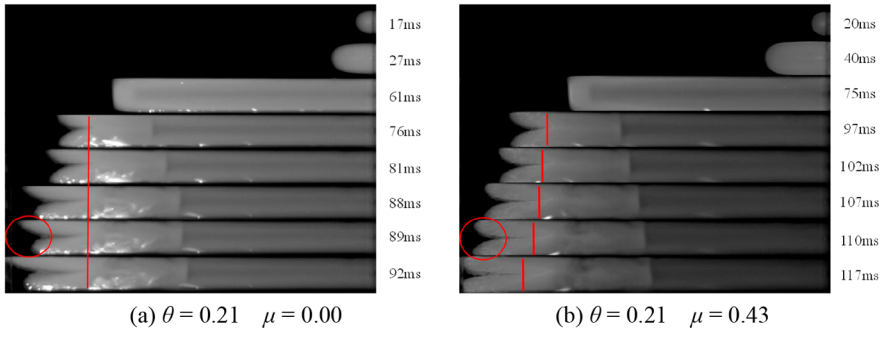


Figure 3. Evolution of tulip flames with (a) $\theta = 0.21$, $\mu = 0.00$ and (b) $\theta = 0.21$, $\mu = 0.43$.

flame does not have sufficient distance to form a flat structure in the duct. At that point, the flat flame is no longer present.

3.2. Flame Tip Dynamics. Figure 6 presents the flame front velocity (V) with various propane fractions μ and oxygen fractions θ. All data come from the same runs as the images of Figure 2. Prior to flame inversion, the flame tip propagating along the duct axis is called the flame front. After the flame inversion, the flame tip near the upper and lower side walls is called the flame front. Z_{tip} is defined as the distance between the flame tip to the ignition position, and V is the time derivative of Z_{tip}.

As shown in Figure 6, the flame front velocity profiles are divided into three patterns. In the first pattern, following initial acceleration, the flame front decelerates sharply and the velocity drops to the minimum. The acceleration process is then repeated again, which can be clearly observed at $\theta = 0.21$. In the second pattern, there is no second acceleration after the flame exhibits acceleration and deceleration, but the deceler-

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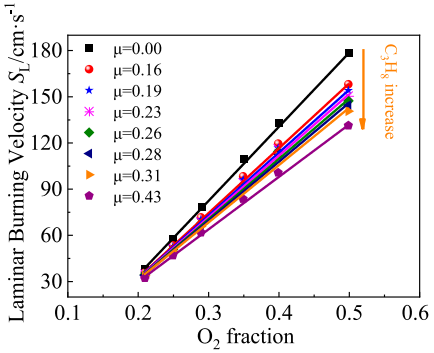


Figure 4. Laminar burning velocities (LBV) computed by the UC San Diego mechanism (the UC San Diego mechanism has been widely used to simulate the combustion of C1−C3 fuels, and it has been verified by Han et al.,[41] Goswami et al.,[42] Li et al.,[43] and Wei et al.[44]).

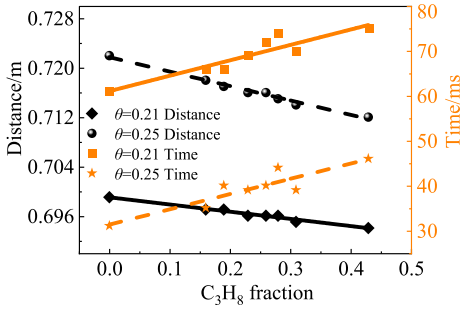
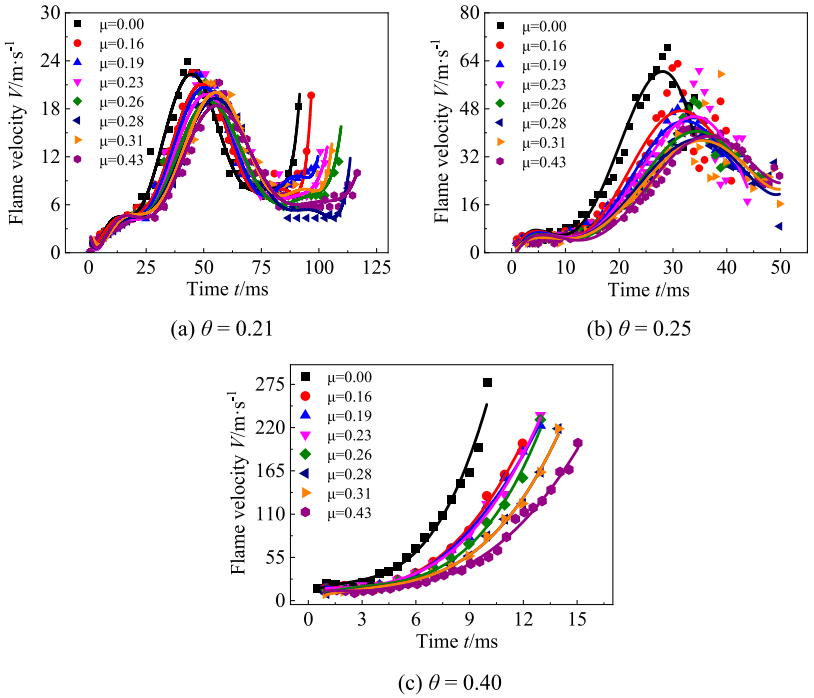


Figure 5. Onset distance and the time of flat flame formation.

ation is gentler, which can be clearly observed at θ = 0.25. In the third pattern, no deceleration can be observed because there is a continuous acceleration process, which can be observed at θ = 0.35, 0.40, and 0.50. Oxygen enrichment leads to the flame front velocity profiles shifting from the first pattern to the second pattern and then to the third pattern.

When the flame structure develops into hemispherical and finger shapes, the flame is in a laminar burning regime. In this regime, the flame surface area increases exponentially,[34,35] which increases the flame propagation and forms the initial flame acceleration. The first deceleration is caused by the flame being partially extinguished by contact with the sidewalls.[34] The flame surface curvature becomes smaller and the flame surface area decreases. Subsequently, the flame front structure reverses and gradually forms a tulip flame, and then the flame front gradually curls and wrinkles, increasing the flame surface area. Next, the reaction intensity and heat release rate increase continuously, leading to the second acceleration. No salient tulip flame can be observed for θ = 0.25. Therefore, the second acceleration is not presented in Figure 6b. As the oxygen fraction increases, the mixture reactivity continues to increase, and the explosion becomes more violent. The quenching effect of the sidewall cannot counteract the flame propagation enhancement caused by this intense combustion, resulting in a continuous increase in flame velocity, as shown in Figure 6c.

Figure 7 presents the propagation time of the flame from ignition to exiting the duct. The red brackets indicate that, as θ increases from 0.21 to 0.25, 0.29, 0.35, 0.40, and 0.50, the decrease in the flame propagation time between adjacent oxygen fractions is 63.1%, 35.3%, 31.8%, 30.1%, and 23.8%, respectively. The difference between 63.1% and 23.8% is significant. Additionally, the black arrows indicate that as μ increases from 0.00 to 0.16, 0.19, 0.23, 0.26, 0.28, 0.31, and 0.43, the decreases in flame propagation time are 21.4%, 32.1%, 31.3%, 25.2%, 36.4%, and 27.3% for the six oxygen fractions, respectively. The difference between 36.4% and 21.4% is relatively small. Compared to propane addition, oxygen enrichment is more effective at reducing the flame propagation time. It is noteworthy that as θ increases from 0.21 to 0.25, the flame propagation time decreases by 63.1%, which



(a) θ = 0.21 (b) θ = 0.25

(c) θ = 0.40

Figure 6. Flame velocity (V) versus time (t) at (a) θ = 0.21, (b) θ = 0.25, and (c) θ = 0.40.

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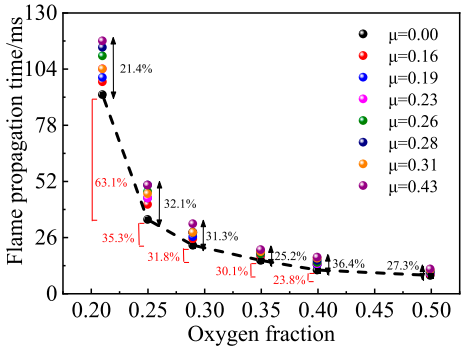


Figure 7. Flame propagation time in the duct.

is much greater than the decreases for the two adjacent oxygen fractions. Therefore, slight oxygen enrichment (i.e., $\theta \leq 0.25$) can effectively reduce the flame propagation time and improve the performance of a mixture.

Figure 8 presents the combined effects of the methane and propane fractions on the maximum flame velocity (V_{max}). V_{max}

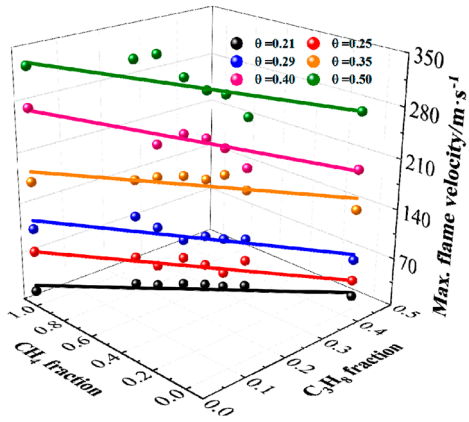


Figure 8. Variations of the maximum flame velocity (V_{max}) with methane and propane fractions.

increases continuously with an increasing methane fraction γ. However, a substantial decrease in V_{max} can be observed with an increase in μ. This contradicts previous observations that propane addition can encourage methane explosions.[17,18,22] The LBV of propane is 1.8 times that of biogas at atmospheric pressure. The heating value of propane is 2.33 times that of pure methane.[16] To ensure a constant fuel heating value according to eq 3, an increase in the propane fraction is accompanied by a decrease in the methane fraction. Additionally, the LBV decreases with increasing the propane fraction (see Figure 4). Therefore, one can see that as the propane fraction increases, the flame propagation time decreases (see Figure 7) and the flame speed decreases (see Figure 8). As μ increases from 0.00 to 0.43, the percentage reductions of V_{max} are 11.3%, 45.9%, 39.6%, 19.1%, 21.4%, and 16.4% for the six oxygen fractions. The inhibitory effect of the propane fraction on V_{max} first increases and then decreases after reaching the maximum inhibitory effect at $\theta = 0.25$. Therefore, the propane fraction significantly decreases V_{max} (especially at $\theta = 0.25$ and 0.29), which reduces the risk of the oxygen-enriched combustion.

3.3. Explosion Pressure Dynamics. As shown in Figure 1, two pressure transducers were employed to monitor the explosion pressure. There are no differences between the pressure curves recorded by the two pressure transducers other than the amplitudes. Therefore, only the pressure at the closed end was analyzed. Figure 9 presents the explosion pressure over time for various propane fractions μ and oxygen fractions θ. There are two types of pressure profiles: unimodal and bimodal. The bimodal overpressure trend appears for (a) θ = 0.21, (b) $\theta = 0.25$, and (c) $\theta = 0.29$. The definitions of P_b, P_{mfa}, and P_{ext} follow those presented by Yang et al. (i.e., P_b is the bursting pressure, P_{mfa} occurs when the maximum flame area is achieved, and P_{ext} occurs as a result of an external explosion).[47] As shown in Figure 9a, where $\mu = 0.23$, P_b is caused by the rupture of the PVC film at the open end. A large amount of unburnt gas is released and the volume expansion rate of initial combustion is much smaller than the rate of the gas escape, resulting in an instantaneous pressure drop. Zheng et al.[38] pointed out that the dynamic failure pressure depends on the reactivity of mixtures, meaning it depends on the combination of μ and θ. Next, an increase in pressure is caused by an increase in the flame area. Finally, a bimodal overpressure trend can be observed.

In most cases, P_b is lower than P_{mfa} in the bimodal pressure trend. P_{mfa} is equal to the maximum peak pressure P_{max}. However, P_b is greater than P_{mfa} when μ is equal to 0.26, 0.28, 0.31, and 0.43 in Figure 9a. At this time, P_b is equal to the maximum peak pressure P_{max}. Additionally, $\mu = 0.26$ is the inflection point of $P_{mfa} > P_b$ in the bimodal pressure trend.

With an increasing θ, the LBV exhibits a substantial increase (see Figure 4), similar to the reactivity of the mixture. The volume expansion rate approaches the rate of gas escape during the initial combustion stage. The rate of pressure drop decreases and the duration of the pressure drop decreases from $\theta = 0.21$ in Figure 9a to $\theta = 0.29$ in Figure 9c. Eventually, the volume expansion rate of initial combustion exceeds the gas escape rate. The slope of the net volume generation rate is always positive, resulting in a continuous increase in pressure, generating a unimodal pressure trend at θ = 0.35, 0.40, and 0.50.

A third pressure transient P_{ext} can be observed in the later stages of the explosion overpressure trend, as shown in Figure 9a. This differs from the damped oscillations highlighted in the previous studies.[38] The time to the local pressure peak P_{ext} is almost the same as the time at which the flame passes out of the duct. Ponizy and Veyssiere,[48] Ferrara et al.,[49] and Di Sarli et al.[50] demonstrated that an external explosion outside a vent duct results in a sudden increase in pressure. P_{ext} can be attributed to the appearance of a small-scale external explosion. The premixed fuel discharged by the explosion in the elongated duct forms a flammable elongated area at the outlet of the duct, which differs from the approximately spherical cloud formed at the outlet of the spherical explosion vessel and the shorter explosion duct. The intensity of the external explosion was not high in this study and it only occurred at θ = 0.21. Additionally, it was also observed that the time of P_{ext} appearance is gradually delayed with increasing propane fraction, as shown in Figure 9a.

Figure 10 presents the variations in P_{max} with the propane fraction μ and oxygen fraction θ with error bars. P_{max} decreases with an increasing μ, which is consistent with the trends observed for V_{max} in Figure 8. As θ increases, P_{max} exhibits rapid growth, similar to the trends of the computed LBV in Figure 4. An increase in the oxygen fraction dramatically enhances the explosion intensity, resulting in greater risk.

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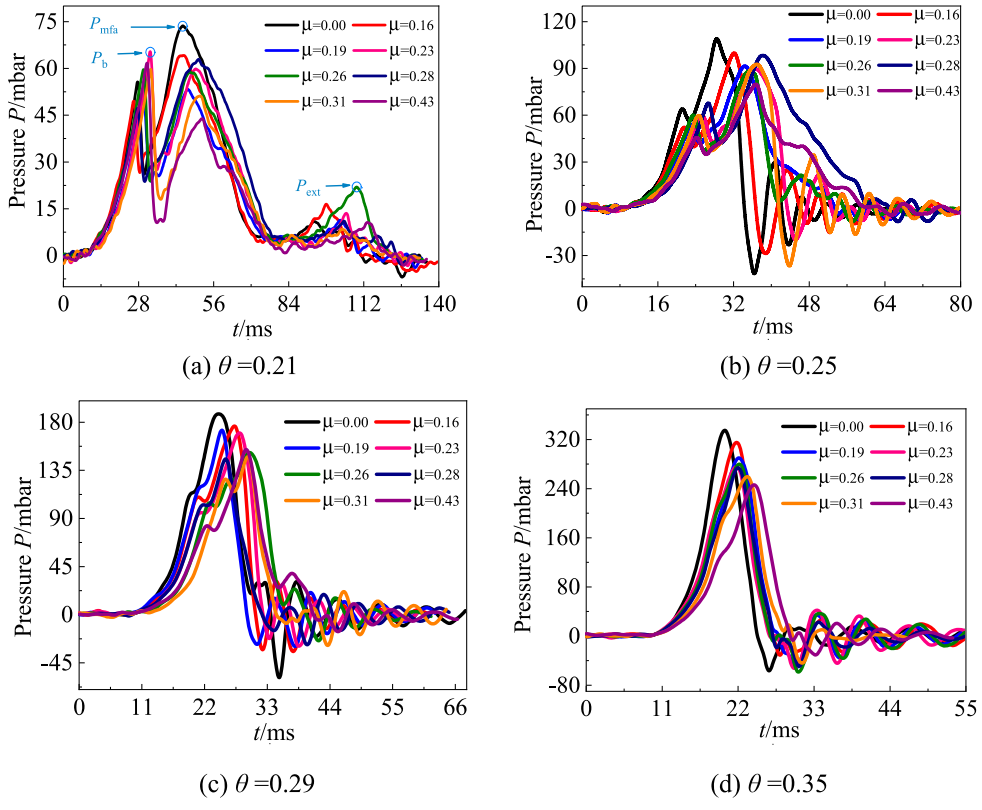


Figure 9. Overpressure history with the oxygen fractions of (a) θ = 0.21, (b) θ = 0.25, (c) θ = 0.29, and (d) θ = 0.35.

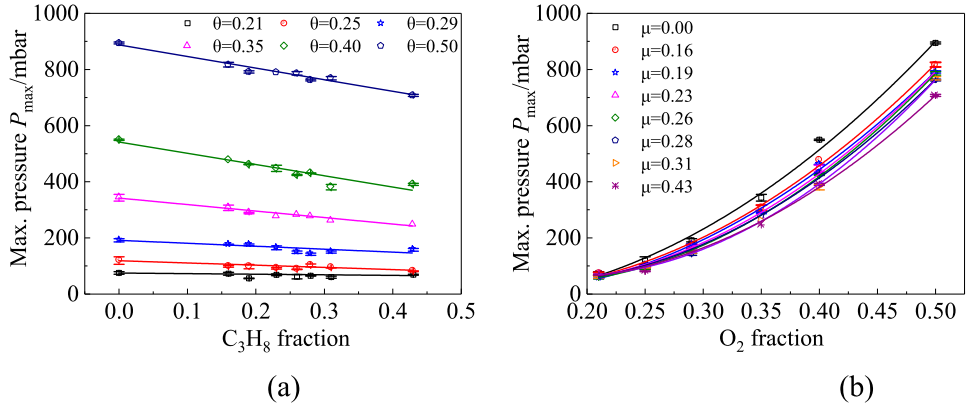


Figure 10. Dependence of the maximum pressure P_{max} on (a) C_3H_8 fraction μ and (b) O_2 fraction θ.

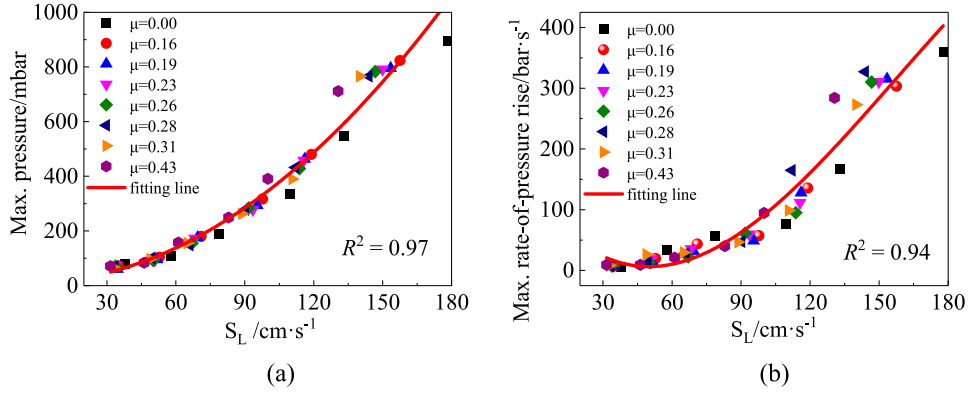


Figure 11. Dependence of the maximum pressure P_{max} on LBV (a) and the maximum rate of pressure rise $(dP/dt)_{max}$ on LBV (b).

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Additionally, as the propane fraction μ increases from 0.00 to 0.43, the percentage reductions of P_{max} at the six oxygen fractions are 14.2%, 18.2%, 17.7%, 26.5%, 29.1% and 26.7%. The inhibitory effects of propane addition on P_{max} first increase and then decrease. This is similar to the conclusions in section 3.2.

Figure 11 presents the variations of P_{max} and $(dP/dt)_{max}$ with LBV. Substantial increases in P_{max} and $(dP/dt)_{max}$ are observed with increasing LBV regardless of the composition of the mixture. P_{max} and $(dP/dt)_{max}$ are important indicators for characterizing the risk and consequences of explosion.[51] Additionally, there is a similar nonlinear trend correlating $(dP/dt)_{max}$ with V_{max} in Figure 12. The greater the V_{max}, the

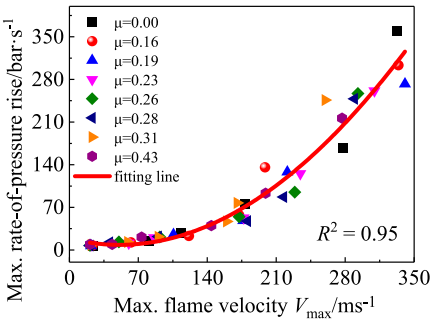


Figure 12. Dependence of the maximum rate of pressure rise $(dP/dt)_{max}$ on the maximum flame velocity V_{max}.

greater the $(dP/dt)_{max}$ and the greater the P_{max}. This also means that there is a higher explosion risk in the use of biogas. Combining the dependence of P_{max}, $(dP/dt)_{max}$, and V_{max} on LBV, the LBV illustrates the combined effect of propane fraction and oxygen fraction and even estimates the explosion risk. However, different geometries have different effects on the propagation of flames, and the correlation shown in Figure 11 is not necessarily appropriate for other configurations.[38,52]

3.4. Rates of Production (ROP) and Radical Distribution. To investigate the effects of propane addition and oxygen enrichment on flame propagation at the elementary reactions level, ROP analysis was performed. As short-lived free radicals in the combustion reaction zone, H, OH, and O are direct participants in elementary reactions, which play an important role in the combustion process.[2,53−56] The primary chain-propagating reactions for producing and consuming H, OH, and O radicals in the mixtures are illustrated in Figure S2 of the Supporting Information (SI). R1 ($H + O_2 \leftrightarrow OH + O$) is the primary chain-propagating reaction that produces the OH and O radicals, but it consumes H radical. R3 ($H_2 + OH \leftrightarrow H_2O + H$) is the dominant chain-propagating reaction that produces the H radical, but it consumes the OH radical. R49 ($CH_3 + O \leftrightarrow CH_2O + H$) is the main chain-propagating reaction that contributes to the reduction of O radicals. Additionally, as μ increases from 0.00 to 0.43 at $\theta = 0.21$, there is an inhibition influence on the ROP of the H, OH, and O radicals. However, as θ slightly increases from 0.21 to 0.25 at $\mu = 0.00$, a significant promotion effect on the ROP of the H, OH, and O radicals can be observed in Figure S2 (SI).

Figure S3 (SI) presents the total ROP for mixtures with (a) $\theta = 0.21$, $\mu = 0.00$, (b) $\theta = 0.21$, $\mu = 0.19$, and (c) $\theta = 0.25$, $\mu = 0.00$. The major species of CH_4, C_3H_8, O_2, CH_3, CO, CO_2, H_2O, and H_2 and the important radicals of H, OH, and O were selected to investigate the effects of propane addition and oxygen enrichment on biogas explosions. The flame temperature (T) increases as the reaction accelerates gradually. At $\theta = 0.21$, when μ increases from 0.00 to 0.19, the maximum total ROP of H, OH, and O decrease from 3.7×10^{-3}, 1.13×10^{-3}, and 4.81×10^{-4} in Figure S3a (SI) to 3.57×10^{-3}, 1.02×10^{-3}, and 4.8×10^{-4} in Figure S3b (SI). The absolute value of the maximum total ROP of O_2 decreases from 1.17×10^{-2} to 1.06×10^{-2}. The addition of propane leads to lower oxygen consumption and less free radicals generation, which reduces the reactivity of the mixture. However, when $\mu = 0.00$, as θ increases from 0.21 to 0.25, the maximum total ROP of H, OH, and O increase from 3.7×10^{-3}, 1.13×10^{-3}, and 4.81×10^{-4} in Figure S3a (SI) to 8.04×10^{-3}, 2.51×10^{-3}, and 1.06×10^{-3} in Figure S3c (SI). The absolute value of the maximum total ROP of O_2 increases from 1.17×10^{-2} to 2.52×10^{-2}. The addition of oxygen leads to greater oxygen consumption and more free radicals generation, which significantly enhances the reactivity of the mixture. Oxygen consumption plays an important role in the entire reaction process.

The radicals H, OH, and O and the species CO, H_2, and CH_3 are converted from production to consumption in the later stages of the reaction process. It is worth noting that the propane consumption rate is faster than the methane consumption rate (Figure S3b, SI). The consumption of propane also ends earlier than the consumption of methane.

The distributions of the H, OH, and O radicals and the variations in the gas temperature at (a) $\theta = 0.21$, $\mu = 0.00$, (b) $\theta = 0.21$, $\mu = 0.43$, and (c) $\theta = 0.25$, $\mu = 0.00$ are shown in Figure S4 (SI). As the reaction begins, the consumption of methane, propane, and oxygen generates a large number of H, OH, and O particles, and the gas temperature increases with distance. The OH radicals occupy the largest fraction and exhibit the slowest decrease. When comparing parts a and b of Figure S4 (SI), as μ increases to 0.43, the mole fractions of the H, OH, and O radicals decrease slightly from 5.4×10^{-3}, 7.9×10^{-3}, and 2.6×10^{-3} to 4.7×10^{-3}, 6.4×10^{-3}, and 2.5×10^{-3}, respectively. However, as θ increases from 0.21 to 0.25, the mole fractions of the H, OH, and O radicals increase significantly by 1.46, 15.57, and 1.65 times, respectively. The greater the mole fractions of the H, OH, and O radicals, the more vigorous the global reaction.[15,55,57] As shown inFigure 13, $[H+OH+O]_{max}$ increases continuously with increasing oxygen fraction. This is consistent with the trend of the LBV versus oxygen fraction in Figure 4. The trend of P_{max} is presented in Figure 10b. Liu et al.[15] pointed out that the LBV exhibits a strong linear relationship with $[H+OH+O]_{max}$. Oxygen enrichment leads to the increase in the microscopic

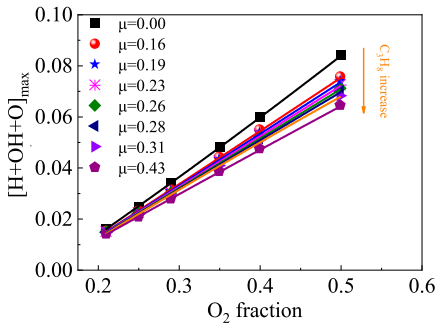


Figure 13. Maximum mole fraction $[H+OH+O]_{max}$ vs O_2 fraction in the oxidizer.

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free radicals concentration and a promotion effect on the ROP of free radicals. The reactivity of the mixture increases accordingly, resulting in an enhanced flame speed and pressure. In contrast, the addition of propane decreased the molar concentrations of microscopic free radicals and inhibited the ROP of the free radicals.

4. CONCLUSION

The effects of propane addition and oxygen enrichment on the flame characteristics of biogas were experimentally studied with a constant heating value in a half-open duct. The flame evolution, flame propagation speed, pressure profiles, maximum pressure peaks, maximum rate of pressure rise, rate of production, and radical distribution were acquired. The main results can be summarized as follows.

Oxygen enrichment has a more significant influence on flame structure evolution compared to propane addition. The evolution of the flame structure revealed that the tulip structure takes on two types of patterns at the oxygen fraction θ = 0.21. As the oxygen fraction θ increases, a salient tulip flame cannot be observed. When θ = 0.21, as the propane fraction μ increases from 0.00 to 0.43, there are different evolutionary processes of the tulip flame and the evolution times of different flame stages are extended. However, when θ > 0.21, there are no obvious changes in the flame structure as μ increases.

The flame front velocity V versus time can be divided into three patterns. The addition of a small amount of oxygen can effectively decrease the flame propagation time and improve the performance of the mixture. Propane addition significantly decreases V_{max} (especially at θ = 0.25 and 0.29) and reduces the explosion risk of the oxygen-enriched mixtures.

Pressure profiles exhibit very different behaviors depending on the combination of propane addition and oxygen enrichment. However, the LBV presents the combined effect of the propane fraction and oxygen fraction, which is an important parameter to predict P_{max}, $(dP/dt)_{max}$, and V_{max} values of the mixtures and estimate the explosion risk.

The rate of propane consumption is greater than the rate of methane consumption, and the consumption of propane ends earlier than that of methane. Oxygen consumption plays an important role in the entire reaction process, leading to the increase in the concentrations of microscopic free radicals and the promotion effect on the ROP of free radicals. In contrast, the addition of propane decreased the molar concentrations of microscopic free radicals and inhibited the ROP of free radicals.

In the future, the effect of the equivalence ratio on the flame characteristics of biogas by propane addition and oxygen enrichment will be studied, and flame accelerations in larger ducts are deserved to be studied for practical applications.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.energyfuels.1c00113.

Tables summarizing the mixture compositions (Table S1) and the chemical kinetic mechanisms and the measurement facilities for laminar burning velocity in literature (Table S2) and figures showing the flame front position (Figure S1); the ROP for radical H, radical OH, and radical O (Figure S2); the total ROP for various mixtures (Figure S3); and effects of the propane and oxygen fractions on the distribution of H, OH, O radicals and gas temperature (Figure S4) (PDF)

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Notes

The authors declare no competing financial interest.

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